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TECHNICAL PAPER

Design and analysis of MEMS based electrospray thruster

K. Srinivasa Rao[1] · Y. Sundar Sai Kumar[1] · K. Sai Sree Rohini[1] · P. Ravi[1] · K. G. Sravani[1] · P. Ashok Kumar[1]

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Abstract

This paper presents a design of MEMS based Electrospray thruster with high throughput, low volume and high output velocities. The performance of thruster can be improved by choosing best possible dimensions of cone height, half angle, propellant height and its radius. The most effective outcomes of the design have been observed through various analysis, which are suggested that propellant height should be > 280 µm, also propellant radius varied between 50 and 125 µm, tip angle with 44°, operating voltage between 1.5 and 3 kV. The elements of the device are designed in micro scale using COMSOL multiphysics tool in Laminar flow environment. Propellant within the emitter usually has ionic liquid, in this design EMI-BF4 is used as propellant. The output velocity is analyzed for input voltages 1.5 kV to 3 kV by varying height and half angle of cone, height and radius of propellant. The optimum outcomes producing the higher efficiencies for the device are achieved for cone angle of 44° with cone height 124 µm and cone bottom radius of 125 µm along with propellant radius of 125 µm and propellant height of 360 µm, the output velocities are of the order 10^6.

1 Introduction

Electrospray thruster is an equipment used for the propulsion of micro-nano satellites i.e. the push is provided to the satellite by using this device inside the satellite. The coalescence of the propellant (Sergio and Parisa 2015) into a Taylor cone and through a charged extraction grid is the basis of propulsion system. Different types of thrusters are available, such as cold gas thrusters, chemical monopropellant systems, pulsed plasma thrusters, vacuum arc thrusters and gun thrusters. Grid ion thrusters and hall effect thrusters offer a high specific impulse among the above systems, but they appear to be larger and require greater power (Levchenko et al. 2018). Electrospray thrusters, which provide 60% to 90% expertise and need a smaller footprint, are easier and highly efficient devices (Kimber et al. 2018). Electrospray thrusters provide an impact of approximately 20 µN. Thrust is produced by accelerating charged particles through electric fields. The charged particles are formed when an electric field is added to a conducting liquid propellant at emitter level (Ziemer 2009).

Previously, the design of thrusters used liquid ion source as a propellant requiring low-heating thrusters while ionic liquids that are now used can be operated at room temperature with fluid metal require high vacuum and emitter preparation temperatures, while ionic fluid emitters can be made at atmospheric pressure (Lozano and Martinez-Sanchez 2005). Ionic liquid has marginal vapor and is ideal to provide sufficient conductivity for electric propulsion (Chiu and Dressler 2007). The performance of the thruster ranges from very high fuel output (species ~ 3000–4000 s in the ion pure regime) to medium-thrusted mode, based on the charge over mass of the released species (~ 1–100 µN per emitter in the droplet phase). The benefits of room temperature and capacity to spray ions of both polarities include the use of ionic liquids in the gas rather than liquid metals (Chakraborty et al. 2012). The important advantage of the electrospray emission system is that the relevant voltage can be either positive or negative in bipolar mode when the high voltage is applied between the propellant and advancing electrode. It has the ability to remove the need for an internal neutralizer with the right choice of ionic liquid (Henning et al. 2018). Thruster usually works through the ionic liquid transferred from the inlet of the propellant in a nozzle which develops in the transmitter upon deformation, following relative increase

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in possible variance, contributing to the formation of the Taylor cone by which droplets are sprayed (Krejci et al. 2016; Coffman 2016). Three different means of transportation can be used for passage of the liquid inside the capillary to the emitter (Lozano and Martinez-Sanchez 2005), wetted through porous material (Robert and Lozano 2011) and wetted internally (Courtney and Lozano 2009; Renato et al. 2008).

In this paper, the latest method is used among the above-mentioned methods because no additional components like pressurized valves for transport of ionic fluid are needed. Internal wetting is often well managed by fluid impedances that would enhance transportation speed and thereby increase device thrust (Dandavino et al. 2013). The use of ionic liquid as propellant, which can even occur in vacuum as a liquid phase, simplifies the propellant supply system and significantly reduces the entire propulsion system (Inoue et al. 2018). Conventional electrospray systems supply ionization liquid through a small capillary to a single transmitter resulting in relatively small output currents and mass flow; emission performance of nanosized droplets or ions, which require absolutely low flow rates, is particularly low (Hill et al. 2014). By varying the design metrics of device which includes height and radius of propellant and height, half angle of Taylor cone, the optimum outputs pertaining to this device is obtained.

In this paper, design of MEMS based electrospray thruster, a propellant and a Taylor cone is designed which is discussed in the first section of paper. Although the formation of Taylor cone is an automatic process, its formation cannot be observed in COMSOL. Therefore, Taylor cone is designed for direct study of the emitter. The active outcomes are observed with several parameters, including the height and radius of propellants and cones respectively, this analysis is demonstrated in the next section which includes all set of dimensions and graphs plotted.

2 Design of proposed MEMS based electrospray thruster

The proposed Electrospray thruster is designed in COMSOL which has propellant and cone called as emitter. Electrospray thruster has ionic liquid in the emitter as propellant. When there is a certain voltage difference created between the emitter and extractor then the ionic liquid is transported from the inlet to the outlet, which results in the thrust which helps in the movement of spacecrafts. In this paper, emitter is analyzed through the output velocity of the thruster i.e. the displacements in the ions ejected from the thruster is observed through simulation. The schematic representation of thruster is mentioned in Fig. 1.

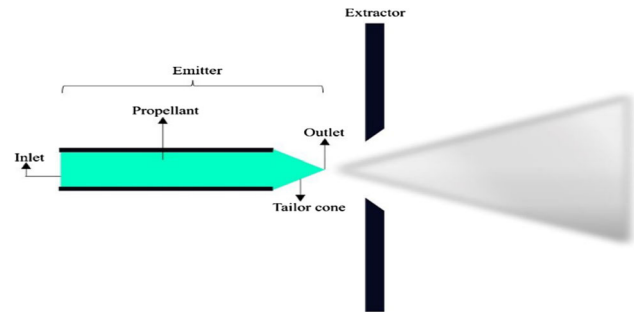

Fig. 1 Schematic view of electrospray thruster

Table 1 Design parameters of electrospray Thruster

Parameter	Dimension
Cone angle	44°
Cone height	124 μm
Propellant height	360 μm
Propellant radius	125 μm

Table 1, illustrates the design metrics of parameters such as propellant radius, cone height, propellant height, cone angle which are analyzed in this paper through which the optimum results are obtained.

In order to create the drive of a spacecraft, material choice plays a prominent role. Ionic liquids conduct fluids electrically, and they have many potential applications as well. The very low vapor pressure of ionic fluids is an improvement to this device and leads to rise in the thrust of the device as well. Here, the device is designed using EMI-BF4 as it is a widely used liquid in ion electrospray with high conductivity properties. The entire mechanism depends on the propellant, so the outer walls of the emitter can be of any material. The emitter has an inlet and outlet through which laminar flow is applied to the device. Among which the inlet has the input velocity applied by correspondingly finding a relation between the velocity and voltage (Ziemer 2009).

$$Ve = \sqrt{2\left(\frac{q}{m}\right)Va}$$

where, Ve : velocity, $\frac{q}{m}$: charge to mass ratio of Ionic Liquid, Va: voltage

The values for varied beam voltages are found out in terms of velocities to be given through inlet and those corresponding values are simulated through the device in the form of normal inflow velocity. Charge to mass ratio of EMI-BF_4 is 523.29 C g^{-1} (Legge et al. 2008). The input voltages are varied from 1.5 kV to 3 kV as the minimum voltage required for formation of Taylor cone if 1.5 kV and if a voltage more than 3 kV is provided, the Taylor cone starts to deform (Alyssa et al.).

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3 Results and discussions

3.1 Design and optimization of Taylor cone

The performance of the Electrospray thruster is associated with the Taylor cone. The Taylor cone is designed in such a way that the thrust produced should be maximum. Hence, the device is optimized such that the thrust produced is maximum. The dimensions of the Taylor cone which includes half angle and height of the cone are varied in the proposed design. The cone half angle is varied from 32° to 44° for the cone heights 35 µm,40 µm,45 µm and 49 µm. Table 2, indicate different output velocities at 32°,36°,40° and 44° for constant cone height of 35 µm.

From Fig. 2, the output velocities for cone half angles is observed in the graphs plotted in Fig. 2 starting from 32° with a step size of 4, the analysis is carried out till the cone half angle reaches 44°. For all the four angles, the voltage vs velocity graph always a linearly raised line. For the initial step the output velocity have started from 4900 m/s and the maximum extent is observed for the 3 kV as 6970 m/s. The same is the case for next step i.e. for 36° for the minimum input voltage the output velocity observed is 7760 m/s and the best output at this half angle is observed as 11,000 m/s for the maximum possible input i.e. at 3000 V. For the next corresponding step the best output velocity produced is 159,00 m/s at 3 kV. The optimum outcomes were observed in the final step i.e. for 44° where there was an incredible increase in the output velocities and produced a maximum output velocity of 26,100 m/s. Table 3, indicate different output velocities at 32°, 36°,40° and 44° for constant cone height of 40 µm.

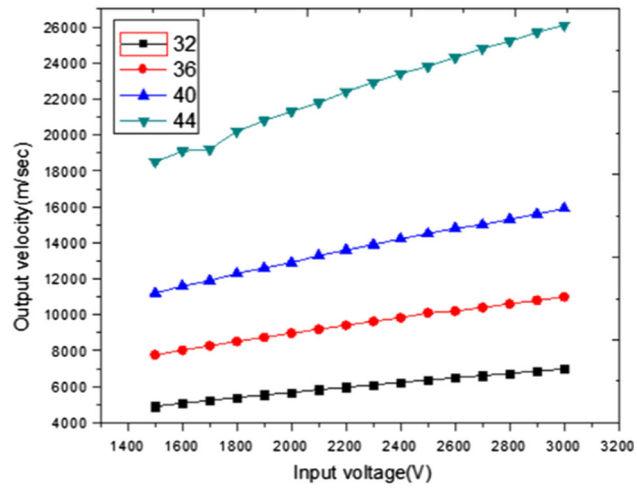

Fig. 2 Input voltages vs output velocities for different cone angles with cone height of 35 µm

From Fig. 3, beginning at 32° and with step size 4, the output velocities for cone half angles are shown until the half angle of the cone is 44°. The voltage-velocity graph always had a linear line for all four angles. For the beginning step, the output velocity is 6690 m/s and for the 3 kV the maximum value is 9510 m/s. The same is true for the following step, i.e. for 36° the output velocity is observed for the minimum input voltage as 10,600 m/s and the best output for the maximal possible input i.e. 3 kV of

Table 2 Output velocities for different cone half angles with cone height of 35 µm

Input voltage (Volt)	Output velocity (m/s)			
	32°	36°	40°	44°
1500	4900	7760	11,200	18,500
1600	5060	8020	11,600	19,100
1700	5220	8270	11,900	19,200
1800	5380	8510	12,300	20,200
1900	5530	8740	12,600	20,800
2000	5670	8970	12,900	21,300
2100	5820	9190	13,300	21,800
2200	5960	9410	13,600	22,400
2300	6090	9620	13,900	22,900
2400	6230	9830	14,200	23,400
2500	6360	10,100	14,500	23,800
2600	6480	10,200	14,800	24,300
2700	6610	10,400	15,000	24,800
2800	6730	10,600	15,300	25,200
2900	6850	10,800	15,600	25,700
3000	6970	11,000	15,900	26,100

Table 3 Output velocities for different cone half angles with cone height of 40 µm

Input voltage (Volt)	Output velocity (m/s)			
	32°	36°	40°	44°
1500	6690	10,600	17,800	35,800
1600	6920	11,000	18,400	37,000
1700	7130	11,300	18,900	38,100
1800	7340	11,600	19,500	39,200
1900	7550	12,000	20,000	40,300
2000	7750	12,300	20,500	41,300
2100	7940	12,600	21,100	42,300
2200	8130	12,900	21,600	43,300
2300	8310	13,200	22,000	44,300
2400	8500	13,500	22,500	45,300
2500	8670	13,700	23,000	46,200
2600	8850	14,000	23,400	47,100
2700	9020	14,300	23,900	48,000
2800	9180	14,500	24,300	48,900
2900	9350	14,800	24,800	49,800
3000	9510	15,100	25,200	50,600

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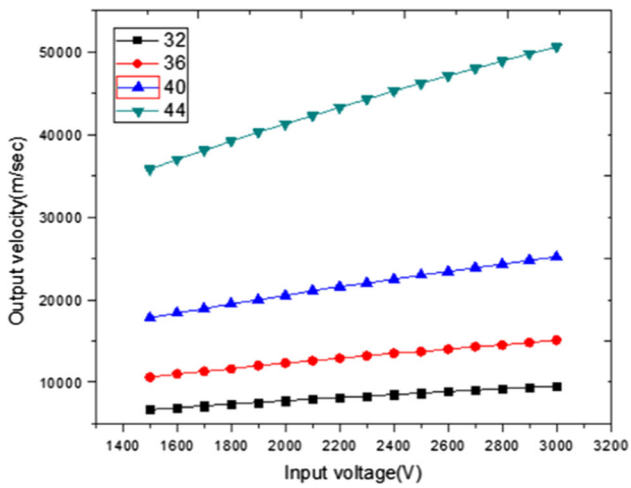

Fig. 3 Input voltages vs output velocities for different cone angles with cone height of 40 µm

15,100 m/s is observed in this half angle. The best performance speed achieved is 25,200 m/s at 3 kV for the next corresponding stage. In the final step, the best effects are seen where the output velocity of 50,600 m/s is observed. Table 4, indicates different output velocities at 32°,36°,40° and 44° for constant cone height of 45 µm.

The performance speed of the cones with half angle is shown in Fig. 4, starting at 32° and with step 4. The voltage vs velocity graph is reliably regular in every four angles. The peak velocity for the first stage is 9780 m/s and the maximum value is for 3 kV i.e. 13,900 m/s. The same graph flow continues in the next step, i.e. for 36° the output

Table 4 Output velocities for different cone half angles with cone height of 45 µm

Input voltage (Volt)	Output velocity (m/s)			
	32°	36°	40°	44°
1500	9780	15,400	31,000	110,000
1600	10,100	15,900	32,000	114,000
1700	10,400	16,400	33,000	117,000
1800	10,700	16,800	33,900	120,000
1900	11,000	17,300	34,800	124,000
2000	11,300	17,800	35,800	127,000
2100	11,600	18,200	36,600	130,000
2200	11,900	18,600	37,500	133,000
2300	12,100	19,000	38,300	136,000
2400	12,400	19,500	39,200	139,000
2500	12,600	19,900	40,000	142,000
2600	12,900	20,200	41,800	145,000
2700	13,100	20,600	41,500	148,000
2800	13,400	21,000	42,300	150,000
2900	13,600	21,400	43,100	151,000
3000	13,900	21,800	43,800	156,000

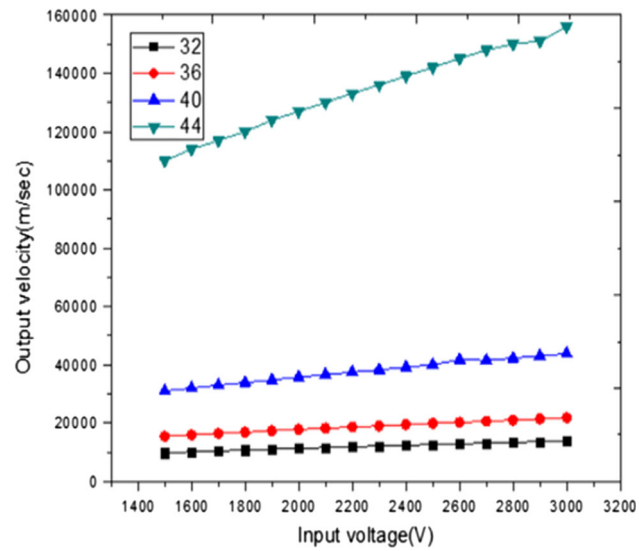

Fig. 4 Input voltages vs output velocities for different cone angles with cone height of 45 µm

value observed for a minimum input voltage is 15,400 m/s. The best output for a maximum input of 3000 V is at 21,800 m/s is shown in this half angle. The best output speed produced is 43,800 m/s at 3 kV for the next corresponding phase. The optimal results are observed in the final stage, when the output speeds increased incredibly and a maximum output speed of 156,000 m/s is produced.

Table 5, indicate different output velocities at 32°,36°,40° and 44° for constant cone height of 49 µm.

Figure 5, indicate output velocities of the half-angled cones starting from 32° with step 4. The voltage vs velocity graph is a linearly increasing line. The first stage peak velocity is 13,300 m/s; for 3 kV velocity is 19,000 m/s. In the next step the same graph stream begins, i.e. the output value measured for 36° with an output velocity of 24,500 m/s In this half angle the highest output is shown with a maximum input of 3000 V at 34,300 m/s. For the next corresponding point, the best output velocity provided is 124,000 m/s with 3 kV. In the final stage the best effects are found with unprecedented changes output velocity achieving a maximum speed of 385,000 m/s.

3.2 Design and optimization of propellant

Propellant height is one of the key parameters of the thruster. Output velocity varies when the propellant height is varied. Holding the cone half angle constant i.e. 44°, the propellant height is varied from 280 µm to 400 µm and outputs are observed for all the input voltages. Below figures are the plots of output velocities for different propellant heights. Outputs are linearly increased for all the propellant heights. The variable output velocities for different propellant heights are indicated i.e. 280 µm,

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Table 5 Output velocities for different cone half angles with cone height of 49 μm

Input voltage (Volt)	Output velocity (m/s)			
	32°	36°	40°	44°
1500	13,300	24,500	71,900	265,000
1600	13,700	25,300	74,200	274,000
1700	14,100	25,800	76,100	280,000
1800	14,600	26,600	77,400	288,000
1900	15,100	27,500	79,200	294,000
2000	15,400	28,300	81,100	306,000
2100	15,900	29,200	83,000	312,000
2200	16,100	29,700	85,200	321,000
2300	16,600	30,200	89,800	329,000
2400	17,000	31,000	90,700	337,000
2500	17,200	31,700	92,800	343,000
2600	17,500	32,300	94,700	349,000
2700	17,900	32,900	96,600	353,000
2800	18,200	33,500	98,200	362,000
2900	18,800	33,800	102,000	373,000
3000	19,000	34,300	124,000	385,000

Table 6 Output velocities for varied propellant heights

Input voltage (KV)	Output velocity 10^6(m/s)						
	280	300	320	340	360	380	400
1.5	1.14	1.1	1.12	1.09	1.14	1.14	1.12
1.6	1.17	1.13	1.15	1.12	1.18	1.18	1.15
1.7	1.21	1.17	1.19	1.16	1.22	1.21	1.19
1.8	1.24	1.2	1.22	1.19	1.25	1.25	1.22
1.9	1.28	1.23	1.26	1.22	1.29	1.28	1.26
2	1.31	1.27	1.29	1.26	1.32	1.32	1.29
2.1	1.34	1.3	1.32	1.29	1.35	1.35	1.32
2.2	1.38	1.33	1.35	1.32	1.38	1.38	1.35
2.3	1.41	1.36	1.38	1.35	1.42	1.41	1.38
2.4	1.44	1.39	1.41	1.38	1.45	1.44	1.41
2.5	1.47	1.42	1.44	1.4	1.48	1.47	1.44
2.6	1.5	1.44	1.47	1.43	1.51	1.5	1.47
2.7	1.52	1.47	1.5	1.46	1.53	1.53	1.5
2.8	1.55	1.5	1.53	1.49	1.56	1.56	1.53
2.9	1.58	1.52	1.55	1.51	1.59	1.58	1.55
3	1.61	1.55	1.58	1.54	1.62	1.61	1.58

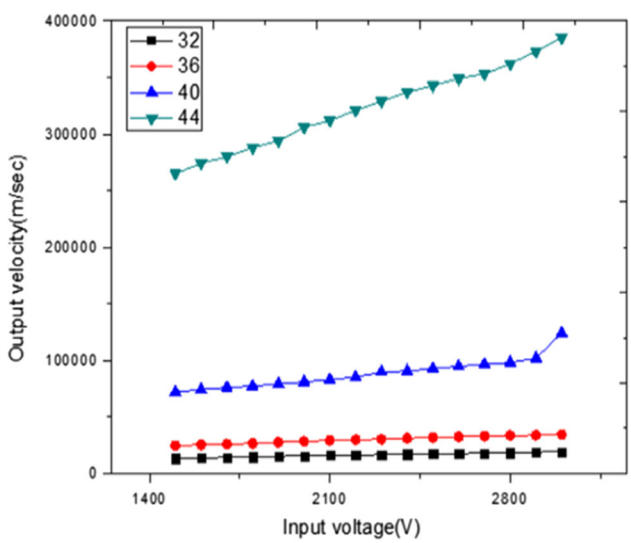

Fig. 5 Input voltages vs output velocities for different cone angles with cone height of 49 μm

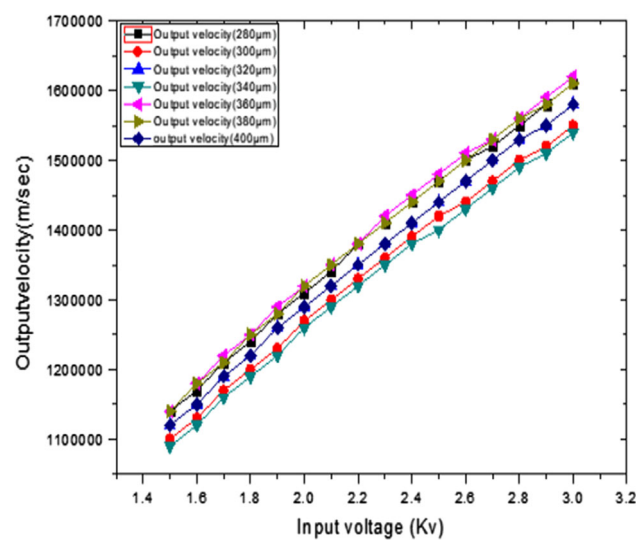

Fig. 6 Input voltages vs output velocities at different cone angles and varied propellant heights

300 μm, 320 μm, 340 μm, 360 μm, 380 μm and 400 μm. Table 6 indicates the output velocities for different propellant heights. Figure 6 shows the corresponding plots.

From Fig. 6, it is observed that output velocities increase linearly for all the propellant heights. The maximum output velocities are observed at 3 kV for all the cases. The output velocities are 1.61×10^6,1.55×10^6,1.58×10^6,1.54×10^6,1.62×10^6, 1.61×10^6, 1.58×10^6 for the propellant heights 280 μm, 300 μm, 320 μm, 340 μm, 360 μm, 380 μm and 400 μm respectively.

3.3 Design and optimization of propellant and cone at the same instance

The height of the propellant and half angle of the cone are kept constant, height of the cone and the radius of the propellant are varied. The propellant radius is varied from

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Table 7 Output velocities for varied cone radius

Input voltage (KV)	Output velocity 10^6(m/s)				
	50	65	80	94	110	125
1.5	1.14	1.46	1.54	1.6	1.7	1.9
1.6	1.18	1.45	1.59	1.65	1.75	1.97
1.7	1.22	1.49	1.64	1.7	1.81	2.03
1.8	1.25	1.54	1.69	1.75	1.86	2.09
1.9	1.29	1.58	1.73	1.8	1.91	2.15
2	1.32	1.62	1.78	1.84	1.96	2.21
2.1	1.35	1.66	1.82	1.89	2.01	2.26
2.2	1.38	1.7	1.86	1.93	2.05	2.31
2.3	1.42	1.74	1.91	1.98	2.1	2.37
2.4	1.45	1.77	1.95	2.02	2.15	2.42
2.5	1.48	1.81	1.99	2.06	2.19	2.47
2.6	1.51	1.84	2.03	2.11	2.23	2.52
2.7	1.53	1.88	2.07	2.14	2.28	2.56
2.8	1.56	1.92	2.1	2.18	2.32	2.61
2.9	1.59	1.95	2.14	2.22	2.36	2.66
3	1.62	1.99	2.18	2.26	2.4	2.7

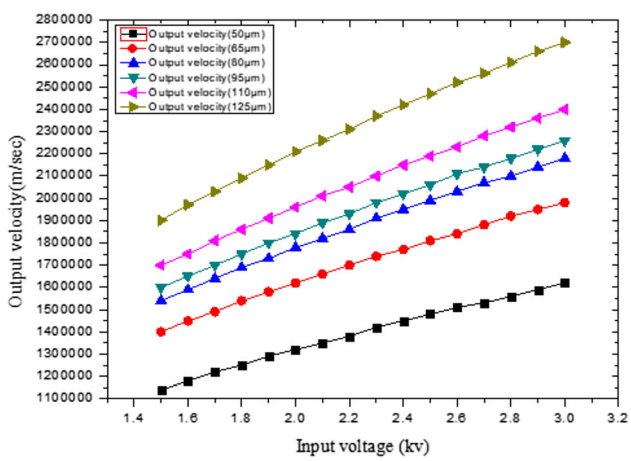

Fig. 7 Input voltages vs output velocities at different cone angles and varied propellant radius

50 μm to 125 μm (Marrese-Reading and Anderson 2016). The cone height is varied accordingly to maintain cone half angle 44°. Changes in output velocities are shown in Table 7 for different cone radii.

From Fig. 7, it is observed that the relative output velocities for corresponding input voltages gives a linearly increasing line for all the propellant radii. It is clearly observed from the graph that there is a huge difference and variance in the output velocities when the propellant radius is correspondingly increased. The minimum and maximum velocities were observed at 1.5 kV and 3 kV respectively.

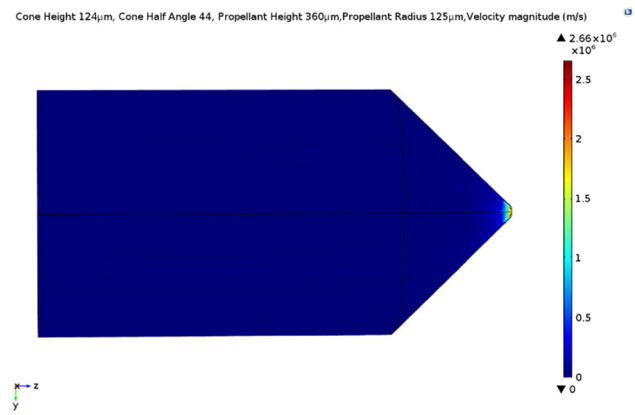

Fig. 8 Schematic view of propellant and Taylor cone designed in COMSOL

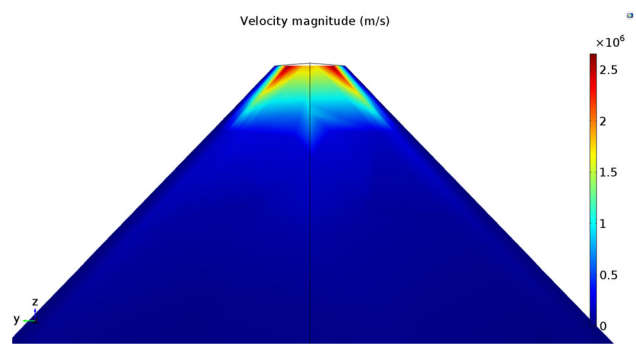

Fig. 9 Schematic view of Taylor cone

The lowest output velocities produced were for 50 μm pertaining to values of 1.14×10^6 and 1.62×10^6 and the maximum output velocity are in the range of 1.9×10^6 and 2.7×10^6 obtained for the propellant radius of 125 μm. Figure 8 shows the schematic of propellant and Taylor cone designed in COMSOL.

The color legend shows the variation of magnitude from the inlet to the outlet. There is an incredible increase in the output velocities when then propellant reaches the tip i.e. Taylor cone. This mechanism is displayed in Fig. 9.

4 Conclusion

In this paper, we have designed MEMS based electrospray thruster. Different design parameters which includes cone height, cone half-angle, propellant radius and height of propellant, play an important role in achieving maximum output velocity. The entire study is carried out in three phases with the Taylor cone, the half angle of the cone ranges from 32° to 44° for the cone heights of 35 μm, 40 μm, 45 μm and 49 μm. The output velocities for 32° and 34° for the input voltages after 2.8 kV are saturated for all four different heights. Maximum output velocity for a

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cone angle of 44° is obtained for the maximum parameters of cone height and cone half angle.

The height of the propellant is calculated where the variable output velocities are evaluated for different heights of the propellant, the maximum output velocities for the height of the propellant 360 µm were observed. Therefore, the thruster model is built with a height of 360 µm. The final phase of the analysis involved in variation of propellant radius from 50 µm to 125 µm, the cone height is evenly varied, so that the results achieved are best and the cone angle 44° is also retained because the previous results were effective. It is noted that for a cone height 124 µm and propellant radius 125 µm the maximum output velocities are obtained.

The proposed design provides the best output velocity of 2.7×10^6 m/s at 3 kV with 124 µm cone height, 44° cone half angle, 360 µm propellant height, and 125 µm propellant radius. The device is suitable for propulsion in nano- and microsatellites when the above-mentioned dimensions can be successfully designed as it delivers maximal output velocities.

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